# SciFig-Evaluation: A Multilingual Benchmark and Behavioural Framework for Evaluating Multimodal Language Models on Scientific Figures

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# Declaration

No portion of the work contained in this document has been submitted in support of an application for a degree or qualification of this or any other university or other institution of learning. All verbatim extracts have been distinguished by quotation marks, and all sources of information have been specifically acknowledged.

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# Abstract

Multimodal large language models are increasingly asked to read scientific figures, yet existing benchmarks largely measure whether a model can extract a value from a chart, not whether it is a faithful visual reader. This thesis introduces SciFig-Eval, a multilingual benchmark of 1,005 figures drawn from 158 arXiv papers and paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, together with a three-axis behavioural framework — Admittance, Resistance and Inductance — that separates honesty in the absence of evidence, faithfulness under misleading context, and sound inductive reasoning. Using an MQM-style multi-judge evaluation pipeline, the framework is exercised on eleven contemporary models spanning five families via five adversarial probe types. The resulting behavioural profile reveals systematic, near-orthogonal failure modes that conventional accuracy scores obscure.

# Acknowledgements

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# Contents

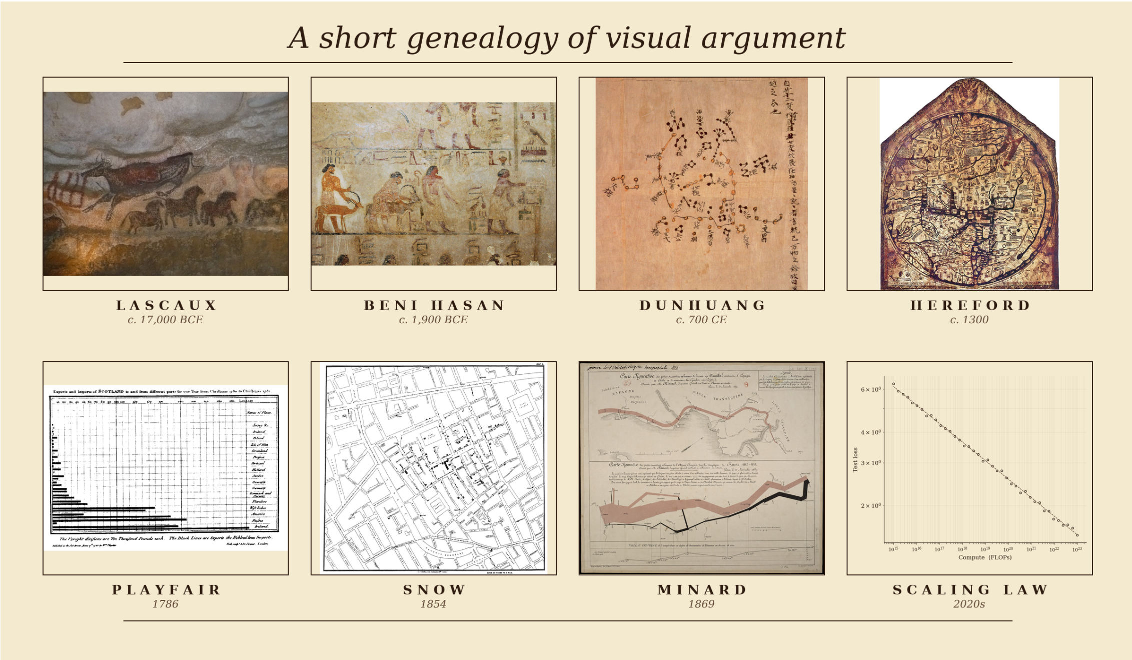
# Chapter 1

# Introduction

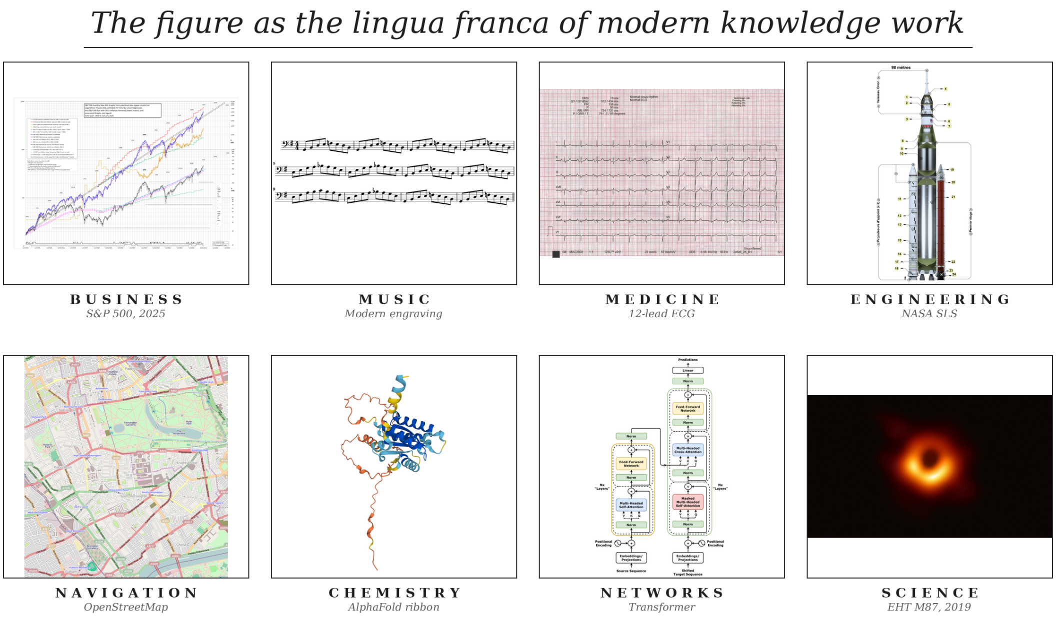
## 1.1 *Homo sapiens*: A Species of Visual Thinkers

The record of a species willing to reason pictorially begins no later than the Upper Palaeolithic. Some time between seventeen and fifteen millennia before the present, an inhabitant of what is now the Dordogne crushed haematite between stones, lit a pine torch, and rendered an aurochs on Lascaux limestone at approximately life size (Aujoulat, 2005). These visual arguments have continued to evolve. Twelfth-dynasty Egyptian painters at Beni Hasan committed a caravan of Semitic travellers, traditionally associated with the Joseph narrative, to a tomb wall around 1900 BCE (Kamrin, 2009); seventh-century Chinese astronomers committed over fourteen hundred stars to silk at Dunhuang (Bonnet-Bidaud et al., 2009); medieval cartographers compressed theology, geography and chronology onto the Hereford *mappa mundi* (Harvey, 1996). In 1786 William Playfair invented the bar chart in order to argue, graphically and in print, about Britain's trade balances (Playfair, 1786). John Snow's 1854 cholera map of Soho laid down the method of modern epidemiology (Johnson, 2006), and Charles Joseph Minard's 1869 flow map of Napoleon's 1812 Russian campaign (Friendly, 2002) is what Tufte (2001) has called, without serious contest, the densest argument ever made on paper (Figure 1.1a). **When a human being intends a claim of any real quantitative weight, the first move is almost invariably to draw.**

In contemporary practice, figures have not only evolved but have become significantly embedded in daily operations. Businesses read through dashboards of candlestick charts and cohort curves (Nison, 2001) that senior management consults in direct preference to prose; engineers coordinate the hundreds of thousands of components that constitute a ship, a hospital or an aircraft through hierarchies of blueprints read without ambiguity by people who may never speak to the designer (Ferguson, 1992); and even life-and-death decisions are made by reading a heartbeat off the grid of an electrocardiogram, or tissue contrast off a radiograph or perfusion map (Goldberger et al., 2017). Music (Treitler, 1984), satellite navigation (Kaplan and Hegarty, 2017), chemistry (Hoffmann, 1995) and network science (Newman, 2010) sustain the same diagrammatic primacy, and every competent empirical paper in the natural and social sciences eventually commits its argument to a figure (Figure 1.1b). **The figure has become a lingua franca of modern knowledge work**, and in many domains the argumentative weight it now carries is strictly greater than the weight carried by the accompanying prose.




**(a)**



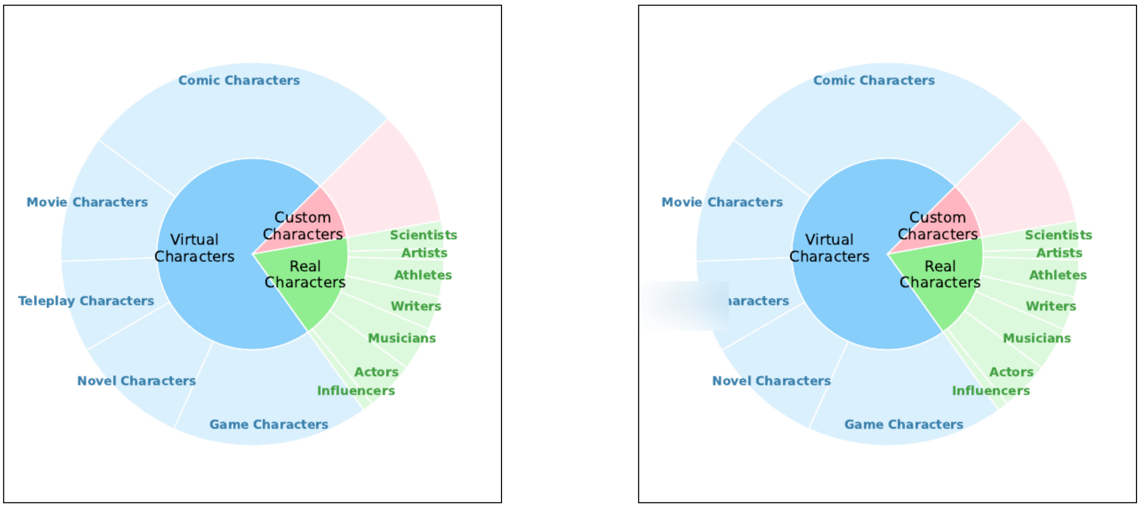

**(b)**

**Figure 1.1:** **(a)** A short genealogy of visual argument, from Lascaux and Egyptian wall painting through the medieval *mappa mundi*, Playfair's 1786 bar chart and Snow's 1854 cholera map to modern neural scaling curves. **(b)** The figure as a lingua franca of modern knowledge work across business, music, medicine, engineering, navigation, chemistry, networks and the natural sciences. *Sources.* **(a)** Lascaux aurochs, Beni Hasan caravan, Dunhuang star chart, Hereford *mappa mundi*, Playfair 1786 bar chart, Snow 1854 cholera map and Minard 1869 flow map of Napoleon's Russian campaign, all public domain (via Wikimedia Commons, the Metropolitan Museum of Art, the British Library, Hereford Cathedral, the Internet Archive and the Bibliothèque nationale de France respectively); scaling curves adapted from Kaplan et al. 2020. **(b)** A candlestick chart, score excerpt, ECG trace, blueprint, nautical chart, molecular diagram, network graph and scientific plot, from public-domain or openly-licensed repositories (Wikimedia Commons, NOAA, PhysioNet) or rendered by the author.

## 1.2 Can *Intelligentia Machinae* (Machine Intelligence) Comprehend Visuals Similarly?

Recent progress in artificial intelligence has rapidly normalised the idea that these systems can be integrated into nearly every part of working life. Having established that *the human* is a visual thinker and that *the figure* is a lingua franca of modern knowledge work, it becomes imperative to investigate how well the current frontier of artificial intelligence, in particular the vision-language models given their present dominance and, looking ahead, world models now emerging alongside them, can navigate the visual paradigm, since in collaborative work with humans they will more often than not encounter images.

There is evidence that the best models of today still fail in some way. Figure 1.2 reproduces a sunburst drawn from an arXiv paper in the SCIFIG-EVAL corpus. In the original image the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in the degraded variant, that single label has been selectively blurred. Asked which category sits between the two, GPT-5.2, arguably the most widely used frontier model at the time of writing, replies "*Anime Characters*" and offers no acknowledgement that the relevant region is unreadable.




**(a)** Original figure. **(b)** One outer label selectively blurred.

**Figure 1.2:** A sunburst from an arXiv paper in the SCIFIG-EVAL corpus. In **(a)**, the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in **(b)**, that single label has been blurred. Asked which category sits between the two, GPT-5.2 answers "*Anime Characters*," with no admission that the relevant region is unreadable. Source image reproduced from arXiv:2505.23923v1 under its open-access licence; blurred variant generated by the SCIFIG-EVAL pipeline.

Beyond cataloguing such gaps in today's VLMs, it is essential to understand their behavioural dynamics under a range of visual stressors and, more importantly, to provide a standardised way of reading those dynamics from their outputs, so that the resulting picture can be used to improve the models' capacity to act as reliable allies in everyday visual work.

## 1.3 Motivation for Focusing on Scientific Research Figures

Among the domains in which collaborative intelligence is now being tested at scale, scientific research has moved fastest. Between late 2024 and early 2026 the three largest model developers each shipped an autonomous research agent (Google, 2024; OpenAI, 2025b; Anthropic, 2025), and scientific deployments such as the Biomni biomedical platform now report research-cycle compressions of an order of magnitude or more (Anthropic, 2026b). Scientific argument is also overwhelmingly visual: every competent empirical paper eventually commits its claim to a figure, so any system that hopes to act as a credible research partner must read those figures with the fidelity its human collaborator does. The scientific figure is therefore the natural object at which to test whether the current frontier reads visually or only appears to.

Science, in addition, is not a monolingual record. A substantial share of the literature is produced in Chinese, German, Bulgarian and other languages of research, and the cost of excluding that share, in citation, uptake and participation, has been documented at length (Amano et al., 2023). The few multilingual figure- understanding evaluations now appearing already report substantial cross-lingual degradation on tasks the same models solve comfortably in English (Li et al., 2025a; Gao et al., 2025; Sharma et al., 2025), which is to say that a figure-reader reliable only in English is, as a research collaborator, structurally partial. The scientific figure considered across languages, then, is what the present work takes as its test object.

## 1.4 Research Questions

The central objective of this thesis is to characterise, across models, languages and adversarial conditions, how faithfully today's frontier multimodal models actually read scientific figures, and to build an evaluation framework that surfaces the failure modes which conventional accuracy benchmarks obscure. Concretely, the investigation is organised around the following four research questions, each pairing a capacity with the conditions under which that capacity is tested:

**RQ1 (Faithful figure description across languages).** To what extent do current vision–language models produce accurate, complete and clearly expressed descriptions of scientific figures across typologically diverse languages?

**RQ2 (Robustness to visual degradation).** How does visual degradation, including noise, compression, aspect distortion and contextual embedding within paper pages, affect the accuracy (whether what is stated is correct), completeness (whether what is visible is reported) and clarity (whether what is reported is expressed unambiguously) of VLM-generated figure descriptions?

**RQ3 (Comprehension beyond description).** Beyond surface description, can VLMs demonstrate genuine comprehension of scientific visualisations, from answering quantitative questions to performing non-trivial inductive reasoning that requires distinguishing analytical insight from surface-level pattern matching?

**RQ4 (Behavioural dynamics under stress).** How do VLMs behave when the visual evidence or its accompanying context is stressed, for example through misleading captions, contradictory premises, sycophancy probes, or images whose answer is no longer pixel-recoverable? In particular, do they hold a faithful reading of the figure, adopt the position the prompt implies, or acknowledge that the available evidence does not support a confident answer?

## 1.5 Research Contributions

In addressing these questions, the thesis makes three contributions.

**C1: SCIFIG-EVAL, a multilingual benchmark for scientific figure understanding.** We release a corpus of 1,005 figures from 158 arXiv papers, paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, and augmented by a 45-figure adversarial subset covering seven image transforms. Figure understanding is scored along three MQM-adapted axes — accuracy, completeness and clarity — through a dual-judge pipeline designed to control for known judge-model biases.

**C2: A psychology-informed adversarial probe design.** Grounded in the misinformation effect and the anchoring bias from cognitive psychology, we construct five probe families — hallucination, prompt-reversal, caption-bias, selective-blur and sycophancy — that target distinct stages of the VLM pipeline, from the visual encoder through cross-modal alignment to the autoregressive decoder, and expose failure modes that conventional accuracy benchmarks conflate.

**C3: The Admittance–Resistance–Inductance behavioural framework.** We propose a three-axis behavioural framework, inspired by the electrical metaphor, that decomposes figure-reading trustworthiness into *Admittance* (honesty in the absence of evidence), *Resistance* (faithfulness under misleading context) and *Inductance* (sound inductive reasoning). Applied across eleven contemporary models from five families, the three axes prove near-orthogonal, recasting trustworthiness as a behavioural profile rather than a single scalar.

## 1.6 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 (Background) provides the technical and conceptual foundation: how frontier multimodal models convert figures into internal representations, how scientific figures carry argument rather than merely depict it, and the failure modes and evaluation traditions on which the remainder draws. Chapter 3 (Literature Survey) situates the work against chart understanding, multimodal hallucination, multilingual evaluation and MQM-style scoring. Chapter 4 (Design) sets out the SciFig corpus, the A–R–L framework and the five adversarial probe families. Chapter 5 (Implementation) describes the evaluation pipeline, the model and judge harness and the reproducibility controls. Chapter 6 (Evaluation) reports the main quantitative results across the 11 evaluated models and the four languages, together with ablations on judge ensembles and probe families. Chapter 7 (Discussion) interprets the findings, connects them to the A–R–L framework and acknowledges the limitations of the study. Chapter 8 (Conclusion) summarises the contributions, revisits the research questions and outlines directions for future work.

## Chapter 2

# Background

This chapter supplies the technical background required to read the rest of the thesis, organised around four questions addressed in turn by §2.1–§2.4.

## 2.1 How do today's Vision-Language Models see?

Contemporary vision-language models share a three-stage pipeline. An image is partitioned into patches and mapped to a sequence of embeddings by a vision encoder; the embeddings are projected by a connector into the token space of a language backbone; and the backbone consumes them as a prefix alongside the text prompt (Liu et al., 2023; Alayrac et al., 2022; Li et al., 2023). By the time the image reaches the language model it is a sequence of soft tokens drawn from the same vector space as text, so every downstream operation on the figure is an operation on these tokens.

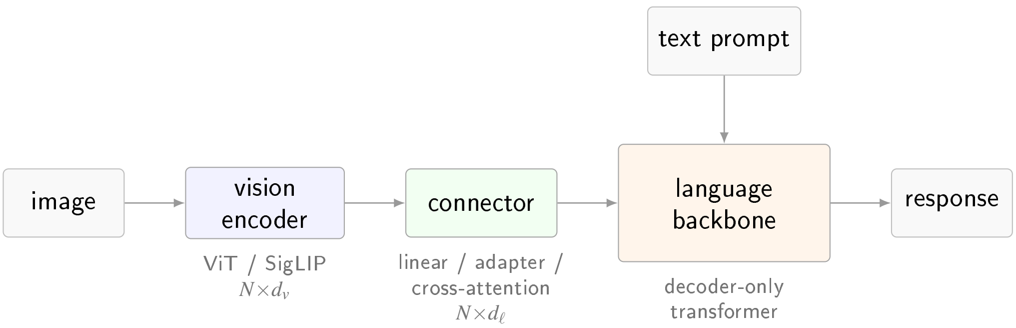



**Figure 2.1:** The canonical vision-language pipeline. A Vision Transformer (Dosovitskiy et al., 2021), typically trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective, maps the image to $N$ patch embeddings of width $d_v$; a connector (linear projection, adapter, or cross-attention module) retunes them to the backbone width $d_\ell$ (Liu et al., 2023; Li et al., 2023); the decoder-only language model consumes them as a prefix alongside the text prompt.

Four architectural choices distinguish frontier families. The *vision encoder* is most commonly a Vision Transformer (Dosovitskiy et al., 2021) trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective; a minority, led by Qwen2-VL and Qwen3-VL (Bai et al., 2024; Qwen Team, Alibaba, 2025), train a custom encoder with 2D rotary position embeddings to accept images at arbitrary resolution. The *connector* is either a linear projection into the backbone's embedding space (Liu et al., 2023), a cross-attention adapter with learnable queries of the Q-Former or Perceiver variety (Alayrac et al., 2022; Li et al., 2023), or a set of LoRA modules (Hu et al., 2022) spliced into the backbone as in Phi-4 multimodal (Microsoft Research,

**Table 2.1:** Publicly documented vision-language architecture across seven representative families spanning the open-weight and closed frontier. "–" denotes a detail that is not disclosed in the primary source. The four open-weight families (top block) publish detailed technical reports; the three closed frontier families (bottom block) publish system cards that address capability and safety but leave the vision pipeline undisclosed. *Sources.* Gemma 3 (Gemma Team, Google DeepMind, 2025); Qwen3-VL (Qwen Team, Alibaba, 2025), with architectural continuity from Qwen2-VL (Bai et al., 2024); Llama 4 (Meta AI, 2025); Phi-4 multimodal (Microsoft Research, 2025); Claude Opus 4.x (Anthropic, 2026a); GPT-5 / 5.2 (OpenAI, 2025a); Gemini 3 Pro (Google DeepMind, 2026).

| Family | Vision encoder | Resolution / tiling | Image tokens | Fusion regime |
|---|---|---|---|---|
| ● Gemma 3 | frozen SigLIP-400M | 896×896 + Pan&Scan | 256 (linear proj.) | adapter |
| ● Qwen3-VL | custom ViT + 2D-RoPE | dynamic, any aspect | variable (MRoPE) | native |
| ● Llama 4 | MetaCLIP-derived ViT | – | – (early-fusion) | native early-fusion |
| ● Phi-4 mm | CLIP variant + LoRA | – | – | Mixture-of-LoRAs |
| ● Claude Opus 4.x | – | $\leq$ 2576 px long edge | – | – |
| ● GPT-5 / 5.2 | – | – | – | native (claimed) |
| ● Gemini 3 Pro | – | up to 900 images / prompt | – | native (sparse MoE) |

2025). The *fusion regime* ranges from a late adapter on a frozen backbone, through early fusion on interleaved tokens (Meta AI, 2025; Chameleon Team, 2024), to native co-training of vision and text from initialisation. The *image-token budget*, finally, varies from a fixed allocation per image (Gemma Team, Google DeepMind, 2025) to aspect-preserving variable-length sequences (Bai et al., 2024), and caps the fine-grained detail the encoder can transmit to the backbone. Table 2.1 summarises where seven representative families sit along these axes.

These axes are instantiated very differently by each family. Qwen3-VL (Qwen Team, Alibaba, 2025), shown in Figure 2.2, is illustrative of the open-weight end of the spectrum, with a distinct vision module (preprocessor, vision blocks with interleaved Multimodal Rotary Position Embeddings, and a patch merger) that injects multi-level features into the language backbone through the DeepStack mechanism and supports the dynamic, any-aspect resolution policy in Table 2.1.

Disclosure along these axes is structurally asymmetric. Open-weight families publish technical reports that document the pipeline end to end, whereas closed frontier families publish only system cards (Anthropic, 2026a; OpenAI, 2025a; Google DeepMind, 2026) that leave the encoder, resolution policy, token budget and connector architecture undisclosed. Any evaluation that would attribute an observed output to a specific pipeline stage by reading internal states therefore has, at the frontier, no internal states to read.

## 2.2 How do today's VLMs reason about what they see, and is it impacted by language?

Reasoning in a vision-language model is autoregressive text generation conditioned on a vision-token prefix. At each decoding step the backbone's self-attention layers attend over a sequence that concatenates projected vision tokens with the text prompt, so every generated token is a function of both channels jointly. Any cognitive operation the model performs on a figure is realised, mechanically, as attention-weighted retrieval from this joint prefix.

Three prompting regimes are layered on this substrate. Under *direct-answer* prompting the

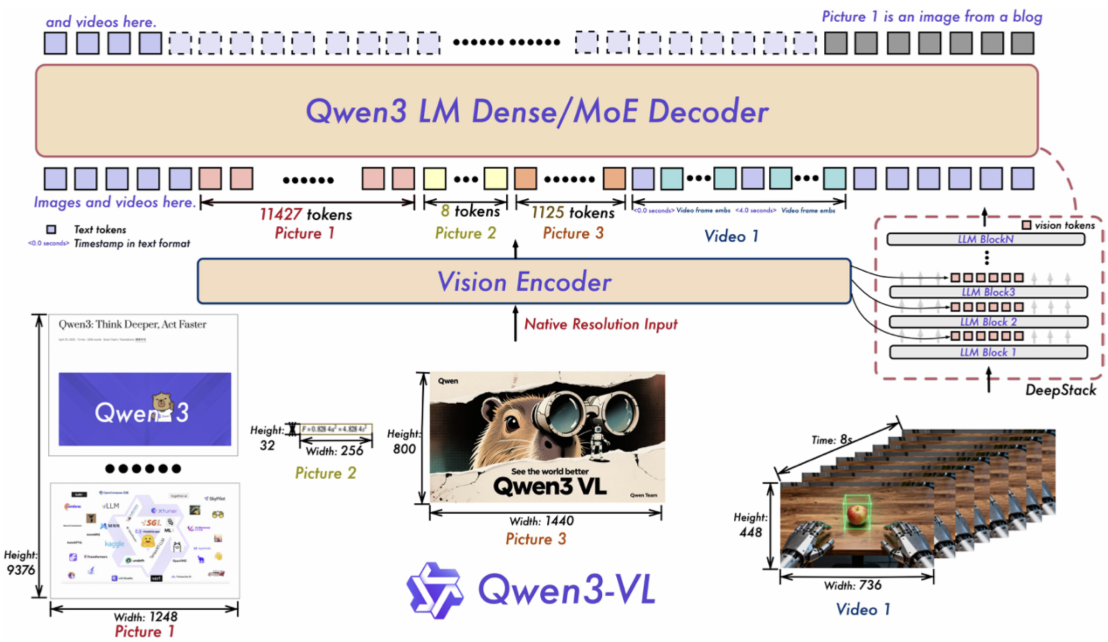



**Figure 2.2:** Qwen3-VL: a distinct vision module (preprocessor, vision blocks with interleaved M-RoPE, and patch merger) injects multi-level features into the language backbone through the DeepStack mechanism. Reproduced from the Qwen3-VL technical report (Qwen Team, Alibaba, 2025).

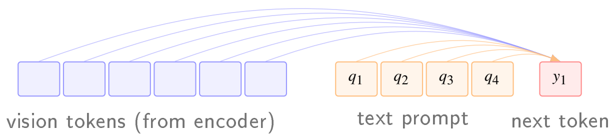



**Figure 2.3:** At each decoding step the next token is produced by self-attention over a single sequence formed by concatenating the projected vision tokens with the text-prompt tokens; cross-modal reasoning is, mechanically, attention-weighted retrieval over this joint prefix.

model emits its answer immediately, and whatever reasoning occurs is implicit in the transformer computation between the prefix and the first output token. Under *chain-of-thought* (CoT) prompting (Wei et al., 2022) the model first emits a natural-language rationale and then the answer, so that each rationale token re-attends to the vision tokens and spreads visual retrieval across more decoding steps. Multimodal extensions couple this loop to image-space scaffolding. Multimodal-CoT (Zhang et al., 2023) interleaves rationale generation with visual features in a two-stage procedure; Set-of-Mark prompting (Yang et al., 2023) overlays enumerated marks on the image so the rationale can reason over discrete regions by reference; and visual chain-of-thought (Shao et al., 2024) conditions the model to emit region crops during reasoning, effectively re-querying the encoder on sub-regions.

Language enters this pipeline at three structurally distinct layers and perturbs reasoning at each. At the *pixel* layer the vision encoder processes non-Latin scripts through the same fixed patch grid it was pretrained on predominantly for English, so fine-grained symbol recognition (Cyrillic diacritics, Chinese logographs at axis-label size, German compound axis labels) degrades before any language-specific reasoning takes place. At the *token* layer the backbone's tokeniser fragments non-English text into finer subword units, lengthening the sequence over which a claim about the figure is represented and diluting the backbone's lexical priors. At the *prior* layer the pretraining distribution is itself English-dominant, so for an underdetermined chart the conditional distribution over continuations is biased toward English-plausible content. The three layers are largely independent and a given perturbation can act on any one or on all three at once.

## 2.3 Behavioural Dynamics of VLMs under Challenging or Adversarial Vision Conditions

The pipeline above does not degrade uniformly when an image is ambiguous, partially obscured, or placed in tension with its prompt. Frontier families differ substantially in how their outputs change under stress, and the range of observed dispositions is wider than a single accuracy number suggests. A model may hold a veridical reading of the figure when a leading caption pulls against it, or quietly revise its description to match the asserted position; this latter pattern is the visual analogue of *sycophancy* (Sharma et al., 2023), the tendency of a language model to adopt an interlocutor's implied position when that position is asserted in the prompt. A model may also combine evidence in the image with domain convention and quantitative reasoning to reach a conclusion the image does not literally state, a capacity often unlocked only under chain-of-thought decoding or the image-space scaffolding of §2.2. And when the pixels required to answer are blurred, occluded, or absent, a model may acknowledge the gap and qualify its answer, or produce a confident substituted answer whose tone is indistinguishable from its answers on unperturbed inputs. The text-only counterpart of that last disposition is the calibration behaviour of Kadavath et al. (2022), who showed that stated confidence in a language model can, under suitable elicitation, be made to track accuracy; a matching property on an adequately perceived image is not guaranteed by the architecture and need not hold when the perceptual prefix is deliberately impoverished.

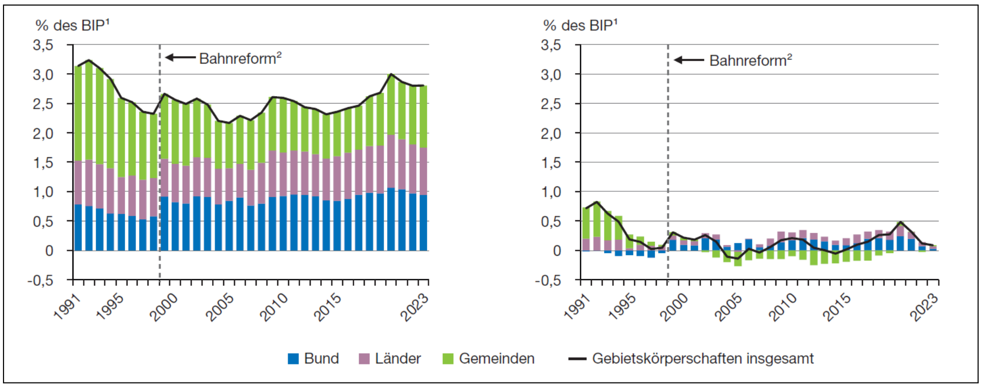
A concrete example makes the phenomenon palpable. Figure 2.4 shows a published German bar-and-line chart of public investment as a percentage of GDP, alongside a variant in which the y-axis ticks, units and legend on the left panel have been blurred while every other visual element, including the right panel's intact axis, remains untouched.

What the empirical literature does not yet supply is a structured way of reading these dispositions off a model's outputs across a controlled stress surface, and that is the gap addressed by the behavioural framework developed in Chapter 4.

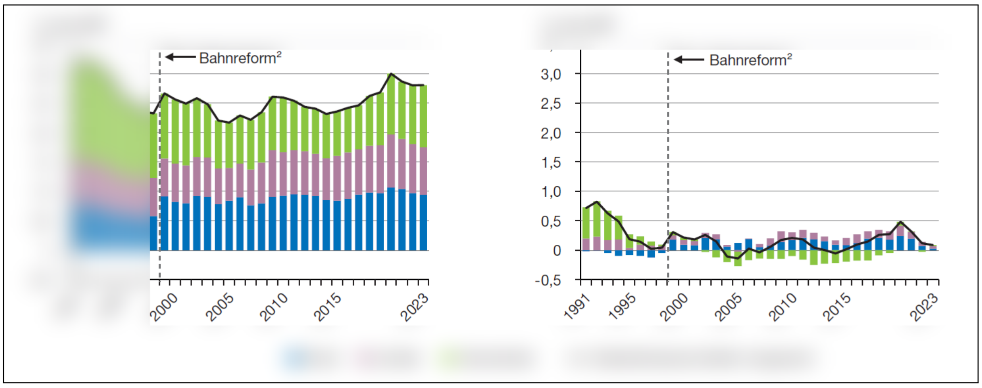
## 2.4 Evaluating VLM Vision Proficiencies

Open-ended visual evaluation of a VLM places two demands on the measurement apparatus: a rubric more expressive than binary accuracy, since the capabilities indexed in §2.3 are not correct-or-incorrect; and a grading procedure that avoids a human annotator for every candidate, since modern multilingual and perturbation-based evaluations routinely yield tens of thousands of outputs.

The reference rubric is Multidimensional Quality Metrics (MQM), introduced by Lommel et al. (2014) for machine translation and since extended across natural-language generation. Each error is tagged with a category from a fixed typology (accuracy, fluency, terminology, style) and a severity weight (minor, major, critical), and a scalar quality score is computed as a weighted




**(a)** Original figure as published.




**(b)** Same figure with the left panel's axis labels, units and legend selectively blurred.

**Figure 2.4:** An axis-blur stress applied to a published German chart of public investment, 1991–2023. **(a)** the original figure, with axis units (*% des BIP*), tick values and legend (*Bund*, *Länder*, *Gemeinden*) all legible; **(b)** the same figure with those elements blurred on the left panel only. The image alone says nothing about how a model should respond; a faithful reader will acknowledge that the left axis is unreadable, an inferential reader may reconstruct it from the intact right panel and the visible gridlines, and a confidently filling reader will invent units and labels that fit the surrounding text. Source: Werding et al. (2025); blurred variant generated by the SCIFIG-EVAL pipeline.

aggregation of the tagged errors. The apparatus transfers to figure description, with the typology becoming visual fidelity, completeness and clarity and the severity weighting distinguishing a misread tick from a fabricated series.

Human MQM annotation does not scale, and the standard alternative is the *LLM-as-judge* paradigm (Zheng et al., 2023), in which a strong general-purpose model grades candidates against a rubric. The paradigm is reproducible and, under careful deployment, reaches human-level agreement on several tasks, but exhibits documented systematic biases: self-preference toward a judge's own family style (Wataoka et al., 2024); position bias that persists across most judges (Zheng et al., 2023); and length and verbosity biases that reward elaboration largely independent of correctness. Randomising candidate position, averaging across position swaps, and aggregating across a heterogeneous judge ensemble form the methodological baseline for open-ended VLM evaluation.

# Chapter 3

# Literature Review

This chapter situates SciFig-Eval against six strands of prior art. Section 3.1 surveys benchmarks for chart and scientific-figure understanding. Section 3.2 reviews hallucination, visual shortcuts and the vision–language bottleneck. Section 3.3 covers robustness, visual degradation and misleading visualisations. Section 3.4 addresses multilingual figure understanding. Section 3.5 treats behavioural evaluation, encompassing honesty, sycophancy and inductive reasoning. Section 3.6 examines the evaluation of evaluators themselves, including MQM, LLM-as-judge and construct validity. Each section closes with the specific gap that SciFig-Eval addresses, and Section 3.7 synthesises the diagnosis that emerges across the strands and positions the present work against them.

## 3.1 Chart and Scientific-Figure Benchmarks

The chart-understanding benchmark literature has arrived in three waves. The earliest, clustered around 2018–2020, is *synthetic*. FigureQA (Kahou et al., 2018) and PlotQA (Methani et al., 2020) generate plots programmatically and ask templated questions about structure, retrieval and numerical reasoning. Their scale is impressive, with PlotQA alone offering close to 29 million question–answer pairs, but their visual diversity is narrow, and models trained on them transfer poorly to real publication figures.

A second wave, beginning with ChartQA (Masry et al., 2022) and extended by Chart-to-Text (Kantharaj et al., 2022), moves to real charts curated from Statista, Pew Research and similar sources, and it broadens the task inventory from closed QA to summarisation. ChartQA's *Relaxed Accuracy* metric, which tolerates small numerical error on open-string answers, set the de facto standard for chart QA, while Chart-to-Text noted early that fluency on chart captions and faithfulness to the chart come apart routinely. The same observation recurs, in stronger form, in CHOCOLATE (Huang et al., 2024a), which analyses factual errors in LVLM-generated chart captions and finds them structured enough to type.

The third wave, 2024–2025, pushes into genuine scientific publication figures and genuine reasoning. CharXiv (Wang et al., 2024) hand-curates 2,323 charts from arXiv and reports that frontier models reach roughly 80% on descriptive items yet only 47% on reasoning, while expert humans remain near 80% on the latter. SciFIBench (Huang et al., 2024b) and ChartQAPro (Masry et al., 2025) sharpen different edges of the same blade. The former targets publication-quality figure interpretation, while the latter introduces unanswerable and conversational items that break

**Table 3.1:** Representative chart and scientific-figure benchmarks in chronological order. *Task* abbreviations cover QA (short-answer question answering), Cap (caption generation), D+R (descriptive and reasoning items scored separately), V/T (visual-primary versus text-primary items) and Desc (paragraph-length description). The *Open* column flags benchmarks whose primary task admits open-ended, paragraph-length generation. The *Reasoning* column flags benchmarks that separate reasoning from description, and the *Behav.* column flags benchmarks that probe behavioural axes such as honesty, abstention or robustness (✓ full, ○ partial, – not supported). SCIFIG-EVAL occupies the intersection that the earlier waves leave unfilled.

| Benchmark | Year | Source | Languages | Task | Open | Reas. | Behav. |
|---|---|---|---|---|---|---|---|
| FigureQA | 2018 | Synthetic | EN | QA | – | – | – |
| PlotQA | 2020 | Synthetic | EN | QA | – | – | – |
| ChartQA | 2022 | Web | EN | QA | – | ○ | – |
| Chart-to-Text | 2022 | Web | EN | Cap | ✓ | – | – |
| CharXiv | 2024 | arXiv | EN | QA (D+R) | – | ✓ | – |
| SciFIBench | 2024 | Scientific | EN | QA | – | ○ | – |
| MultiChartQA | 2024 | Real (web) | EN | Multi-chart QA | – | ✓ | – |
| ChartMuseum | 2025 | Real (web) | EN | QA (V/T) | – | ✓ | – |
| ChartQAPro | 2025 | Web + news | EN | QA (conv./unans.) | ○ | ✓ | ○ |
| PolyChartQA | 2025 | Re-rendered | EN + 9 | QA | – | ○ | – |
| **SCIFIG-EVAL** | **2026** | **arXiv** | **EN/BG/DE/ZH** | **Desc. + probes** | ✓ | ✓ | ✓ |

saturation on easier chart benchmarks. ChartMuseum (Tang et al., 2025) labels questions by reasoning type and exposes a large gap between *text-primary* items, on which models look competent, and *visual-primary* items, on which they collapse. MultiChartQA (Zhu et al., 2025) adds a compositional axis, requiring cross-chart integration that single-chart benchmarks cannot capture.

Three properties characterise this benchmark literature in aggregate. First, it is overwhelmingly short-form. Systematic evaluation of open-ended, paragraph-length descriptions of real scientific figures remains thin, with Chart-to-Text and CHOCOLATE the main exceptions. Second, it is overwhelmingly English. Only a handful of the benchmarks cited above include non-English splits, and none cover typologically diverse scientific figures at publication quality. Third, reasoning questions rarely separate “reasoning the figure compels” from “reasoning the caption invites”, a distinction the next section will show to be central. SCIFIG-EVAL extends this strand along the three axes on which it is thinnest, namely open-ended, multilingual, reasoning-aware description of real publication figures, scored through typed error dimensions rather than exact match.

## 3.2 Hallucination, Visual Shortcuts, and the Vision–Language Bottleneck

Hallucination on chart-like inputs has been documented along three complementary lines of attack. The first treats hallucination taxonomically. ChartHal (Xu et al., 2025) organises fine-grained hallucination on chart QA into irrelevant, non-existent and contradictory scenarios and shows that strong models remain poor once each scenario is scored separately. CHOCOLATE (Huang et al., 2024a) performs the equivalent exercise on chart *captioning* and finds that fluent captions systematically smuggle in factual errors. HallusionBench (Guan et al., 2024) further decomposes hallucination into two distinct failure modes. The first, *language hallucination*, occurs when the decoder over-generates beyond what vision supports. The second, *visual illusion*, occurs when the encoder misperceives. HallusionBench argues that treating the two jointly has obscured both.

The second line asks whether the image matters at all. Li et al. (2025b) run a striking diagnostic. On popular visualisation-QA benchmarks, many items can be solved at frontier-model accuracy *without showing the image*, because either question context or option structure leaks the answer. The implication is that much of what is reported as multimodal competence is in fact parametric recall from the language side. ARO (Yuksekgonul et al., 2023) shows, in parallel, that several vision–language models behave like bags of words on captioned images, under-using order and relational structure in ways consistent with a textual rather than visual solution. ChartMuseum's (Tang et al., 2025) sharp gap between text-primary and visual-primary items, at similar aggregate difficulty, is the same phenomenon from the evaluation side, in that apparent chart competence often reflects reading chart *text* rather than reading chart *graphics*.

The third line attempts a shallow mechanistic account. FUGU (Tartaglini et al., 2025) runs controlled scatter-plot interventions on three vision–language models and argues that a major bottleneck sits at the vision–language hand-off. Correct coordinates can be probed out of the vision tower while the final answer is wrong, and supplying oracle coordinates helps single-point tasks yet sometimes hurts aggregate ones. Fine-tuning alone does not reach ceiling, which points to a residual architectural limit rather than a data-coverage issue.

These three lines jointly explain why purely accuracy-based benchmarks flatter models. The language decoder can manufacture plausible output, the evaluation protocol can reward it, and the image itself is often redundant to the extent that text context leaks. SCIFIG-EVAL inherits the ChartHal and CHOCOLATE error typology but applies it to *open-ended descriptions* of real scientific figures in four languages, and it builds explicit probes, including a selective-blur family that removes specific information from the image while preserving the question, to separate "cannot see" from "will not admit".

## 3.3 Robustness, Degradation, and Misleading Visualisations

A parallel literature asks whether clean-chart competence survives realistic imperfection and adversarial design. Two strands are directly relevant.

**Robustness to image noise and prompt reframing.** CHAOS (Moured et al., 2025) perturbs chart tasks along visual and textual axes and documents that multimodal models are systematically sensitive to realistic perturbation. "Losing the Plot" (Shin et al., 2025) pairs corruption and occlusion with harder multiple-choice distractors and adds a *prompt-reverse* self-consistency check, in which the model is asked first which candidates are correct and then which are incorrect, that reveals a distinct failure class beyond sampling noise. Ahn and Yin (2025) formalise that class as prompt-reverse inconsistency. EncQA (Ng et al., 2025) slices the perceptual axis differently, organising chart QA by visual encoding channel (position, length, colour, shape) and showing that difficulty is strongly channel-dependent, so that models are not uniformly "good at charts" in any meaningful sense.

**Misleading design and visualisation literacy.** A closely related literature asks whether models can *spot* deception. Visualisation-literacy work (Pandey and Ottley, 2025) evaluates vision–language models on standard instruments (VLAT) and their critical counterparts (CALVI), and finds that models that look reasonable on basic literacy items fail on misleading-design items humans flag easily. Misviz (Tonglet et al., 2025) builds a taxonomy-typed detection benchmark at large scale, and the "Perils of Chart Deception" study (The Perils of Chart Deception, 2025) enumerates specific deceptive design patterns (truncated axes, dual scales, aspect distortion) and shows models tracking the same cues that mislead human viewers.

What is missing across both strands is a unified *behavioural* profile. Robustness studies measure accuracy or MQM drops under degradation, while misleading-chart studies measure detection rates on adversarial renderings. Neither tells us how the *same model* responds to both stressors simultaneously, nor whether its failure under each is principled abstention, over-confident fabrication or capitulation to misleading context. SCIFIG-EVAL applies both classes of stressor (seven image transforms and five adversarial probe families) to the same figures, and scores the resulting behaviour along three near-orthogonal dispositions rather than reducing it to a single scalar.

## 3.4 Multilingual Figure Understanding

The multilingual axis is the youngest of the strands surveyed here. PolyChartQA (Li et al., 2025a) decomposes English seed charts into structured JSON plus rendering code, translates the text layer into nine additional languages and re-renders, and reports a substantial English-versus-other-language gap that instruction tuning narrows but does not close. Figure 3.1 shows eight representative instances drawn from their release, illustrating how the chart layout and underlying data are held fixed across languages while only the text layer is translated and re-typeset. PM4Bench (Gao et al., 2025) is a parallel-multilingual suite spanning OCR, document, knowledge and chart tasks across ten languages, and reaches similar conclusions. Strong frontier models degrade sharply cross-lingually, and the gap widens for smaller open models. IndicVision (Sharma et al., 2025) extends the diagnosis to Indic languages specifically.

Two limitations characterise this strand. First, it is dominated by short-form QA rather than open-ended description, so cross-lingual degradation on *generation quality* is under-documented. Second, the figures themselves are either synthetic or re-rendered from English seeds, while *native* figures drawn from non-English scientific papers, paired with expert annotations in the paper's own language, remain rare. The broader case for this gap is made by Amano et al. (2023), who document the measurable costs of excluding non-English research in citation, uptake and participation. SCIFIG-EVAL samples figures from genuinely multilingual arXiv papers in English, Bulgarian, German and Chinese, and pairs them with expert annotations in the source language, closing this particular gap.

## 3.5 Behavioural Evaluation of Honesty, Sycophancy and Induction

Beyond accuracy, a third literature evaluates the *dispositions* of large language and vision–language models, asking whether they admit what they do not know, whether they resist pressure that contradicts the evidence, and whether they can infer legitimately from partial information.

**Honesty and abstention.** BeHonest (Chern et al., 2024) organises honesty into self-knowledge, non-deceptiveness and consistency, and includes *expressing unknowns* and *admitting knowns* scenarios that probe each dimension separately. The abstention survey (Wen et al., 2025) catalogues metrics (ER, C@Acc, AUROC, AUACC, AURCC) and the accuracy–coverage trade-offs they encode. SimpleQA (Wei et al., 2024) introduces a correct / incorrect / not attempted three-way grading that rewards appropriate abstention rather than penalising it, and TruthfulQA (Lin et al.,

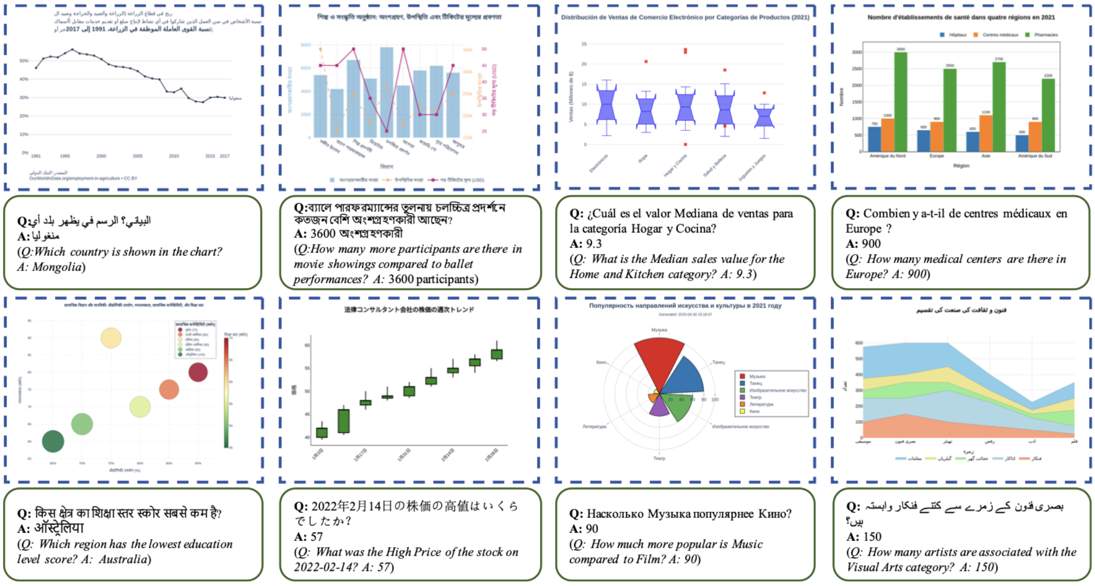



**Figure 3.1:** Representative multilingual chart question-answering visualisations selected from PolyChartQA. Top row, from left to right, Arabic, Bengali, Spanish and French. Bottom row, from left to right, Hindi, Japanese, Russian and Urdu. The underlying data and chart layout are held fixed across languages, while the text layer (title, axis labels, tick labels, legend and the accompanying question and answer) is translated and re-rendered. Image reproduced from Li et al. (2025a).

2022) measures the opposite failure mode, namely the tendency to echo common human misconceptions. Ren et al. (2024) argue, conversely, that over-abstention is itself a failure mode, and propose evidence-gathering procedures that reduce unnecessary refusals on vision–language tasks. Earlier, Kadavath et al. (2022) showed that large language models are capable of surprisingly well-calibrated self-evaluation on text tasks, providing a calibration reference point for the behavioural measures above.

**Sycophancy and evidence grounding.** Sharma et al. (2023) demonstrate that RLHF-trained models systematically adopt the users' stated positions across free-form tasks, and trace the effect to human preference signals favouring agreeable responses, and Zhao et al. (2024) extend the diagnosis to vision–language models specifically. SycEval (Fanous and Goldberg, 2025) distinguishes *progressive* sycophancy, in which pressure happens to lead to correct answers, from *regressive* sycophancy, in which pressure overrides correct ones, and tests both via single-turn rebuttal probes. Sycophancy under Pressure (2025) extend the setup to multi-turn adversarial dialogues in scientific QA. Most tellingly for the present work, Evidence Grounding Under Pressure (2026) ablate evidence composition and uncertainty cues across nineteen instruction-tuned models and find that evidence alone does not reliably prevent user-aligned reversals, so misleading context can override ostensibly correct premises even in frontier models. ELEPHANT (ELEPHANT, 2025) extends the sycophancy lens to *social* face-preservation, reporting that large language models preserve user face roughly forty-five percentage points more often than humans on identical scenarios.

**Inductive reasoning from partial information.** When some of the evidence is missing, perhaps a blurred axis label or a covered legend, the relevant question is no longer "can you see?" but "can you infer from what is still visible?" MIRAGE Team (2025) evaluate inductive reasoning in language models with meta-rules and meta-facts, and find reliance on neighbour-based reasoning rather than correct rule inference. This is the cognitive register in which a blurred axis demands not a fabricated value but a principled inference bounded by what the visible data plausibly permit.

**Psychological priors.** These behavioural failure modes have well-documented analogues in human cognition. The anchoring-and-adjustment heuristic (Tversky and Kahneman, 1974) describes how an irrelevant numeric anchor biases subsequent quantitative estimates, and the misinformation effect (Loftus and Palmer, 1974) shows how a single leading suggestion presented after an event contaminates memory for the event itself. The adversarial probes used in SCIFIG-EVAL are deliberately designed in this register. A misleading caption anchors the reader, a contradictory premise plants a misinformation seed, and both are checked against what the figure itself actually shows.

Collectively, this literature supplies the vocabulary (admittance, resistance, induction) that SCIFIG-EVAL's three-axis framework formalises on *visual* evidence specifically. The originality of the present work is not in introducing any of these notions, each of which has substantial prior art, but in operationalising them simultaneously on scientific figures, across languages, under typed and judge-audited scoring.

## 3.6 Evaluating the Evaluators through MQM, LLM Judges and Construct Validity

Any evaluation framework is only as good as its measuring instrument. Three bodies of methodological work inform SCIFIG-EVAL's instrumentation.

**MQM and structured rubrics.** The Multidimensional Quality Metrics framework (Lommel et al., 2014) was developed for machine translation evaluation and specifies error typologies, severity tiers and rater guidelines. Its adaptation to visual captioning is still young, with CHOCOLATE (Huang et al., 2024a) the closest precedent, applying error typing to chart captions. SCIFIG-EVAL adopts the MQM skeleton (typed errors with severities) but adapts the quality dimensions to *accuracy*, *completeness* and *clarity*, the three axes that matter most for a scientific-figure description.

**Reference-based and reference-free metrics.** Text-generation evaluation began with n-gram overlap (BLEU, ROUGE), migrated to contextual-embedding similarity with BERTScore (Zhang et al., 2020), and extended to cross-modal reference-free scoring with CLIPScore (Hessel et al., 2021). For factuality specifically, FActScore (Min et al., 2023) introduced the decompose-then-verify paradigm. Each generation is broken into atomic facts, every fact is verified against a knowledge source, and the proportion supported is reported. CHAIR (Rohrbach et al., 2018) operationalises the same idea for object hallucination in image captioning. The decomposition-and-check logic transfers directly to figure descriptions, where each atomic claim can be checked against what the figure actually shows.

**LLM-as-judge and its biases.** The LLM-as-judge paradigm (Zheng et al., 2023) scales open-ended evaluation substantially, but comes with documented biases such as self-preference (Wataoka et al., 2024), position bias and length or verbosity bias, all of which an unregulated judge propagates. Comprehensive surveys (LLMs-as-Judges Survey, 2024) catalogue inter-rater agreement measures (Cohen's and Fleiss' Kappa, percent agreement) and trust-calibration methods, and Judge's Verdict (2025) prescribe a two-step correlation-then-kappa methodology for validating judges against human raters. OLAF (OLAF, 2025) formalises LLM-based annotation as a measurement process with explicit treatment of reliability, calibration, drift, consensus and aggregation.

**Construct validity.** At the outermost layer, "Measurement to Meaning" (Salaudeen et al., 2025) distinguishes *criterion validity*, where the measured quantity is directly observable, from *construct validity*, where the quantity is a latent trait operationalised through observables, and adapts psychometric validity facets to AI evaluation. "Measuring What Matters" (Burnell et al., 2025) reviews 445 LLM benchmarks, finds pervasive operationalisation weakness, and offers an actionable checklist. These frameworks motivate explicit construct-mapping in SCIFIG-EVAL. Admittance, resistance and inductance are treated as latent constructs, operationalised through probe families designed so that observed probe behaviour is a faithful indicator of the underlying disposition.

## 3.7 Synthesis and Positioning

The five substantive strands above converge on a shared diagnosis. Frontier multimodal models often appear competent on clean English single-chart QA while, under closer examination, they (i) lean on textual shortcuts whenever the language side affords an answer, (ii) fabricate fluent content when visual evidence is thin or removed, (iii) capitulate to misleading context that a faithful reader would resist, and (iv) fail cross-lingually on tasks they solve comfortably in English. The methodological strand (§3.6) then makes clear that accuracy scores alone obscure these failure modes and that construct validity becomes non-trivial once latent behavioural dispositions enter the rubric.

No existing benchmark currently measures these failure modes *simultaneously* on scientific figures, across languages, with an evaluator-of-evaluators discipline. SCIFIG-EVAL is positioned precisely at that intersection. Its corpus closes the multilingual-scientific-figure gap of §3.4. Its MQM-adapted, open-ended scoring closes the open-description-at-scale gap of §3.1. Its adversarial probe families jointly stress the hallucination and shortcut modes of §3.2 and the robustness and deception modes of §3.3. Its admittance–resistance–inductance framework formalises the behavioural axes that the honesty, sycophancy and induction literatures converge on (§3.5). Its judge-ensemble, kappa-audited pipeline is designed against the construct-validity and judge-bias concerns of §3.6. Chapter 4 develops the dataset and probe design in full, Chapter 5 describes the evaluation pipeline, and Chapter 6 reports the quantitative results across the evaluated models and languages.

**Table 3.2:** Gap map. Each row is a failure mode established in the literature. The middle column lists representative works that isolate the mode in pure form, and the right column names the SCIFIG-EVAL probe family and the behavioural axis (**A**dmittance, **R**esistance or **I**nductance) under which the same mode is stressed on real scientific figures, across four languages, simultaneously.

| **Failure mode** | **Prior work isolating it** | **SCIFIG-EVAL probe / axis** |
|---|---|---|
| Textual shortcut and image-redundant answering | See-or-Recall (Li et al., 2025b), ARO (Yuksekgonul et al., 2023), ChartMuseum visual/text gap (Tang et al., 2025) | No-image and no-caption ablations, selective-blur probes (R) |
| Fabrication under thin visual evidence | ChartHal (Xu et al., 2025), CHOCOLATE (Huang et al., 2024a), HallusionBench (Guan et al., 2024) | Inexistent-entity, unanswerable and irrelevant probes scored for refusal (A) |
| Capitulation to misleading context | SycEval (Fanous and Goldberg, 2025), evidence-grounding ablations (Evidence Grounding Under Pressure, 2026), Sharma et al. (2023) | Wrong-caption, contradictory-premise and prompt-reverse probes (R) |
| Cross-lingual brittleness | PolyChartQA (Li et al., 2025a), PM4Bench (Gao et al., 2025), IndicVision (Sharma et al., 2025) | Native EN / BG / DE / ZH figures with source-language expert annotations (all axes) |
| Deception blindness on misleading design | CALVI (Pandey and Ottley, 2025), Misviz (Tonglet et al., 2025), Perils of Chart Deception (The Perils of Chart Deception, 2025) | Misleading-chart subset scored for miss and false-alarm rates (I) |

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